

AI531C — T07 · The Measurement Studio: Student Sheet

Tue 15 Sep, 2:00–3:00 PM · run: python3 T07_studio.py (+ 'support' / 'extension')

Your mission

Turn a binary mask into MEASUREMENTS — area, centroid, aspect, extent — and discover the structuring-element size gate.

Experiment

Work through steps [1]–[3]: SE sweep 5/25/35/45 → survivors 24/23/2/2 (the gate slams near coin diameter); measurement table of 5 largest — the impostor band shows aspect 4.20, extent 0.42 vs coins' $\sim 1.05/0.77$; median coin area 1492 px.

Vary / break it

Between SE 25 and 35 nearly everything dies. Binary-search the critical size; relate it to the median coin diameter ($\sqrt{(4 \cdot 1492/\pi)} \approx 44$).

Discuss

Q1 · The impostor's extent is 0.42 vs ~ 0.77 for coins. Define extent and explain the gap.

Q2 · Why does the SE gate close abruptly rather than gradually as SE size grows?

Q3 · Area-filtering (stats) and opening (morphology) can both remove small objects. When do they differ?

Reflect

This table is the ancestor of every lane's results table: objects → numbers → decisions.

Fast? EXTENSION cell: circularity $4\pi A/P^2$ per object; find the least circular and explain from its patch. Struggling? SUPPORT cell: read the provided table only; circle the column that separates coins from the band. Nothing is due after class — the hour is the work.